DATA SCIENCE CAPSTONE PROJECT

SpaceX Falcon 9
First-Stage Landing Prediction

Exploratory analysis • SQL • Folium • Plotly Dash • Classification

Project repository

https://github.com/MatiasMontenegrom/spacex-falcon9-landing-prediction

Evidence-based revision | Screenshots and notebook outputs supplied for the project

Executive summary

What this project delivers and what the evidence supports

- Combines launch-record exploration, SQL aggregation, geographic mapping, an interactive dashboard, and supervised classification.
- The dashboard filters launch site and payload-mass range, then compares payload mass with launch outcome and booster version.
- SQL identifies four launch-site labels in the source and returns 619,967 kg as the sum of recorded payload masses.
- Historical model outputs show Decision Tree test accuracy of 94.4% (17 of 18 test examples); this is a small test set, so treat it as preliminary.

4

Distinct site labels

619,967 kg

Total recorded payload mass

94.4%

Decision Tree test accuracy*

18

Test observations*

*Cached notebook result; re-run for final reproducibility.

Introduction & objective

Why predict Falcon 9 first-stage landing success?

- Reusable first stages can reduce launch costs when recovery is successful.
- The project asks whether recorded launch attributes help predict the binary landing outcome.
- Target used in the course dataset: Class (0/1). Predictors include payload mass, launch site, orbit, flight number, and related features.
- Prediction is an analytical aid—not a guarantee of a future landing.

01 **Collect records**

02 **Clean & explore**

03 **Map & query**

04 **Train classifiers**

05 **Evaluate**

Data collection & wrangling

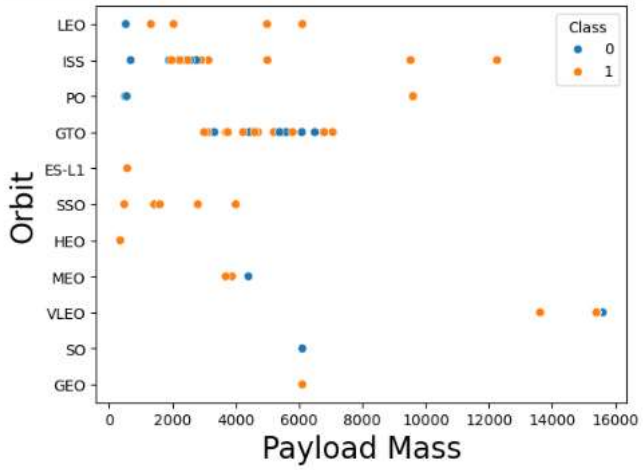
Data fields, transformations, and quality considerations

TASK 5: Visualize the relationship between Payload Mass and Orbit type

Similarly, we can plot the Payload Mass vs. Orbit scatter point charts to reveal the relationship between Payload Mass and Orbit type

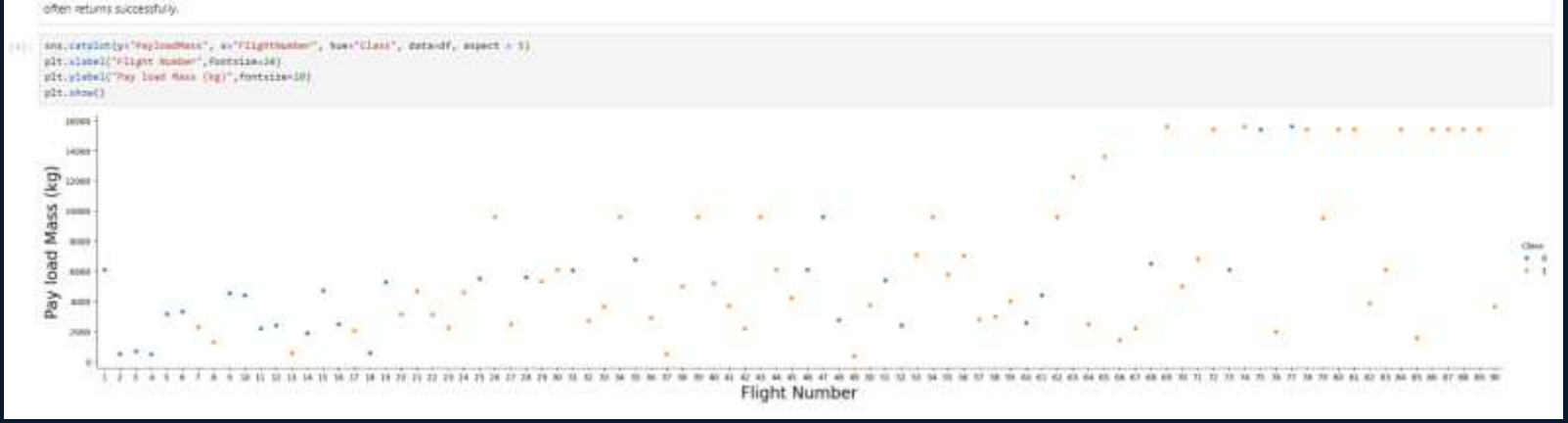
```
# Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value

sns.scatterplot(y="Orbit", x="PayloadMass", hue="Class", data=df)
plt.xlabel("Payload Mass",fontsize=20)
plt.ylabel("Orbit",fontsize=20)
plt.show()
```



EDA overview: payload mass by flight

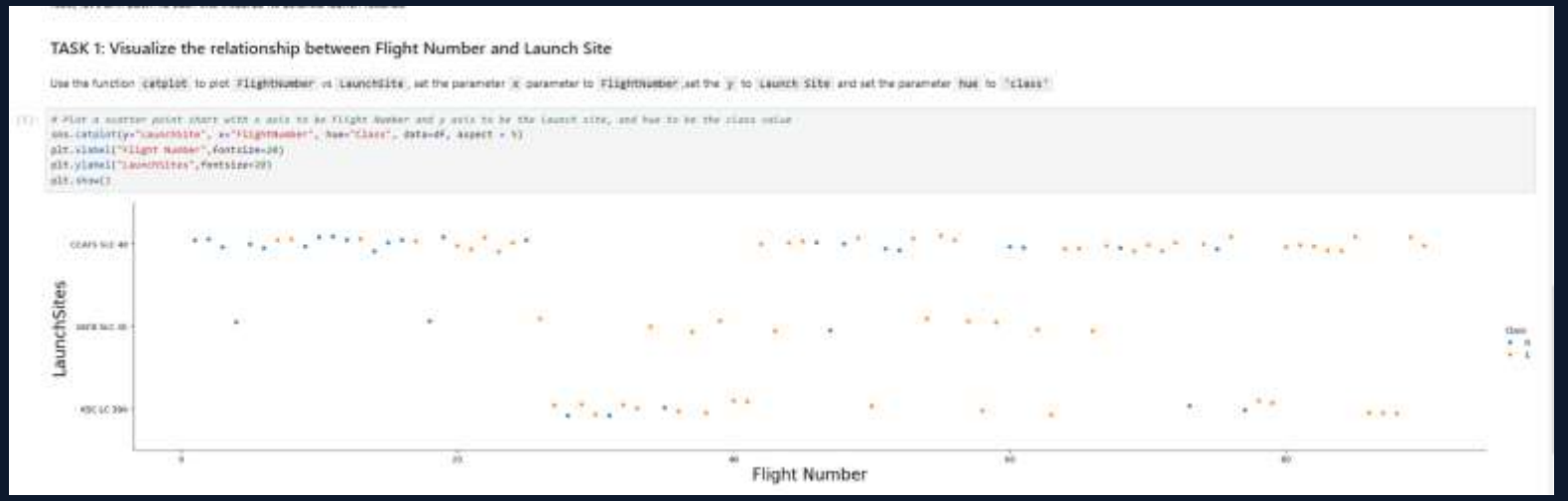
The supplied plot shows a wide range of payload masses and mixed outcomes



- Payload mass varies from very small values to above 15,000 kg.
- Both classes occur across the flight sequence; payload mass alone does not cleanly separate outcomes.
- High-mass missions are present among both successful and unsuccessful outcomes.

EDA: launch site and flight number

Success and failure labels are distributed across multiple sites



- CCAFS SLC-40 has the densest concentration of records in this plot.
- Later flight numbers include more observations at KSC LC-39A and VAFB SLC-4E.
- Because sites have different sample sizes and mission mixes, raw counts should not be interpreted as site-specific causal effects.

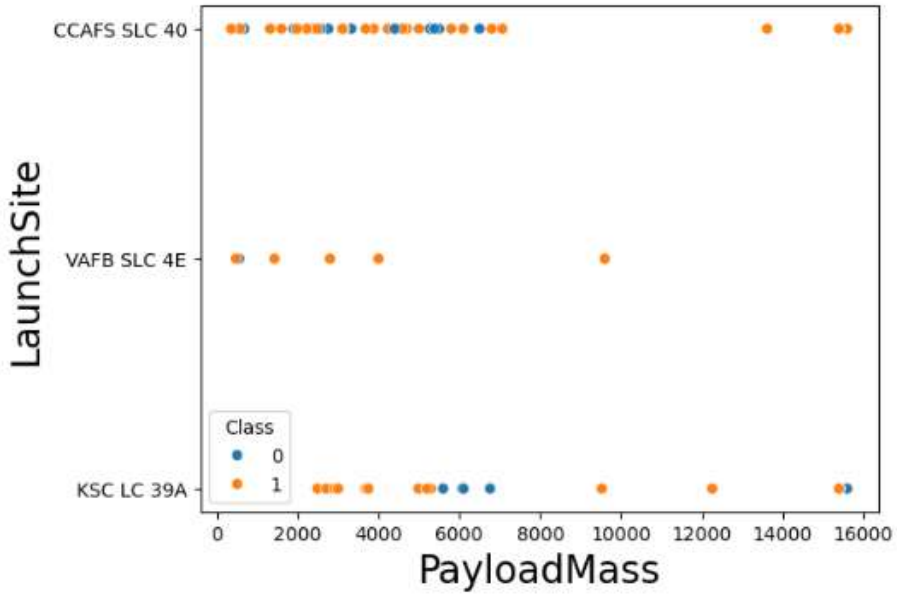
EDA: payload mass by launch site

Payload distributions differ by site in the supplied sample

TASK 2: Visualize the relationship between Payload Mass and Launch Site

We also want to observe if there is any relationship between launch sites and their payload mass.

```
[6]: # Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
sns.scatterplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df)
plt.xlabel("PayloadMass",fontsize=20)
plt.ylabel("LaunchSite",fontsize=20)
plt.show()
```



Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than

- The VAFB SLC-4E observations shown do not include payloads above 10,000 kg.
- CCAFS SLC-40 and KSC LC-39A include a broader spread of payload masses.
- Compare rates only after accounting for the number and type of missions at each site.

EDA: observed success rate by orbit

Some orbit categories reach 100% in this sample, but sample sizes matter

Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

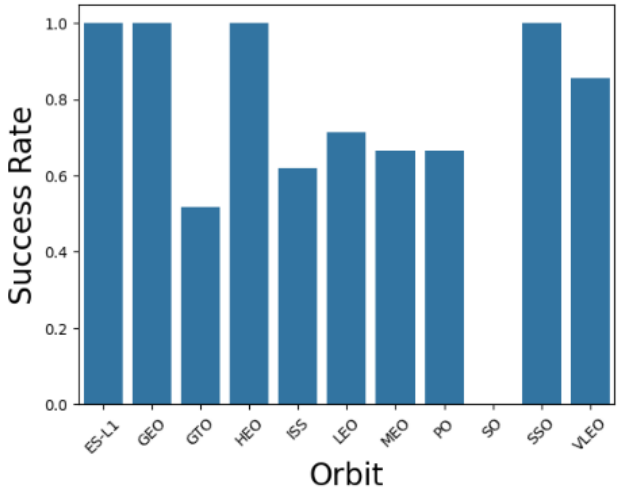
TASK 3: Visualize the relationship between success rate of each orbit type

Next, we want to visually check if there are any relationship between success rate and orbit type.

Let's create a `bar chart` for the sucess rate of each orbit

```
orbit_success = df.groupby('Orbit')['Class'].mean().reset_index()

sns.barplot(data=orbit_success, x='Orbit', y='Class')
plt.xlabel("Orbit", fontsize=20)
plt.ylabel("Success Rate", fontsize=20)
plt.xticks(rotation=45)
plt.show()
```



Analyze the plotted bar chart to identify which orbits have the highest success rates.

- ES-L1, GEO, HEO, and SSO appear at 100% in the chart.
- GTO is near 52%; several other categories fall between roughly 60% and 86%.
- SO has no visible bar, likely because no usable observations appear for that category.
- These are sample proportions, not stable population estimates.

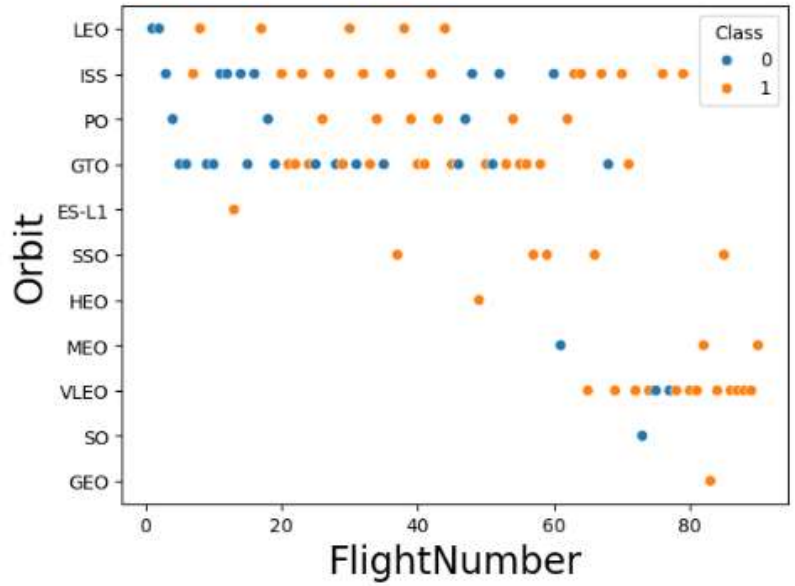
EDA: flight number and orbit type

Flight number appears to co-vary with the types of missions launched

TASK 4: Visualize the relationship between FlightNumber and Orbit type

For each orbit, we want to see if there is any relationship between FlightNumber and Orbit type.

```
[9]: # Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.scatterplot(y="Orbit", x="FlightNumber", hue="Class", data=df)
plt.xlabel("FlightNumber",fontsize=20)
plt.ylabel("Orbit",fontsize=20)
plt.show()
```

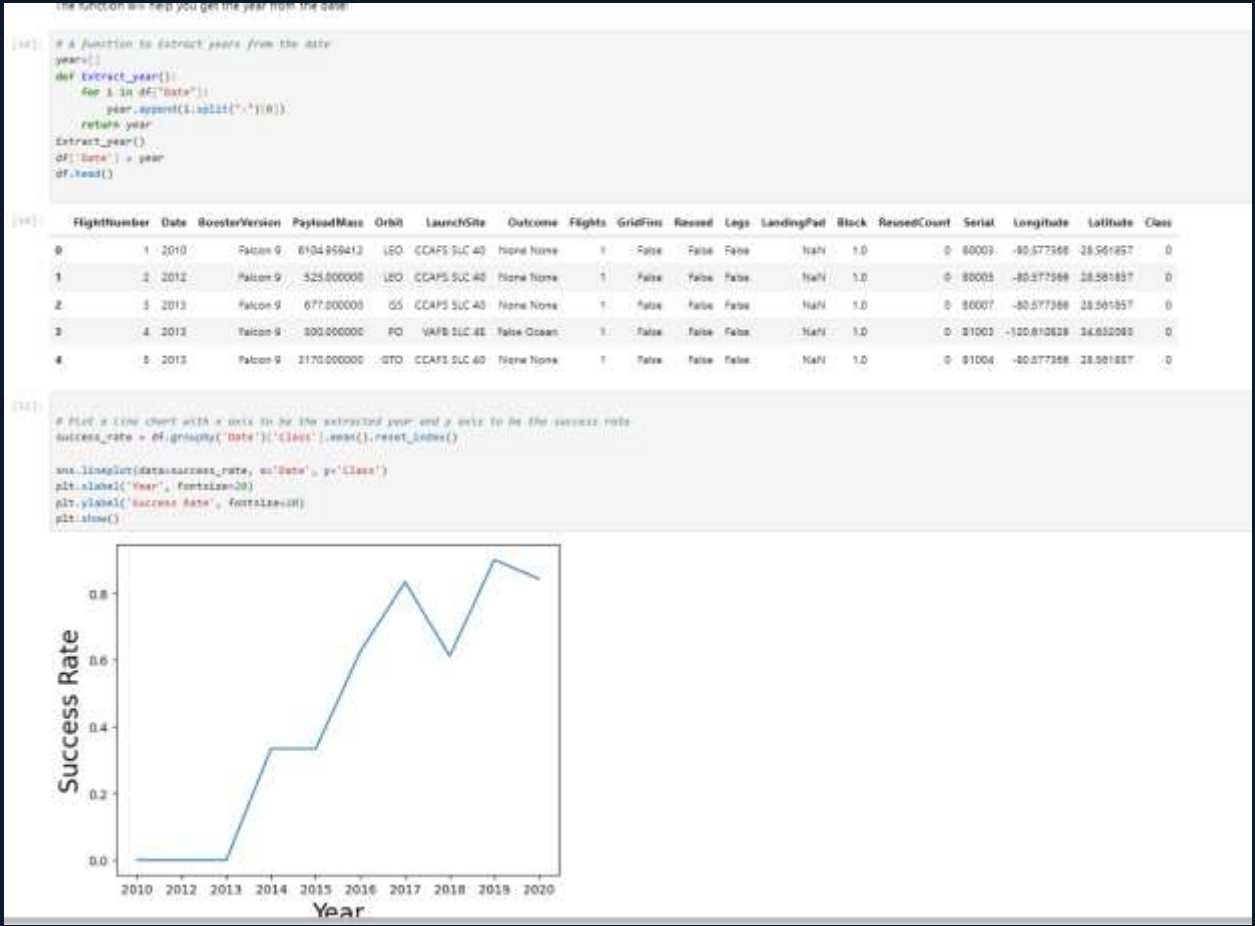


You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight numbe

- LEO and GTO observations span much of the flight sequence.
- VLEO appears concentrated in later flight numbers in this dataset.
- Mission mix changes over time, so a model may learn temporal or mission-type patterns as well as landing-related signals.

EDA: annual success-rate trend

The plotted annual rate rises after the early years



- The supplied chart shows near-zero rates in the earliest years, then a rise from 2014 onward.
- The line reaches approximately 0.83 in 2017, dips to about 0.61 in 2018, and peaks near 0.90 in 2019.
- Early years have very few observations; annual averages can swing sharply when the denominator is small.
- Treat the trend as descriptive, not proof that year itself causes success.

EDA with SQL: query-backed findings

Selected results from the executed SQLite notebook

SQL question	Observed result
Distinct launch-site labels	CCAFS LC-40; VAFB SLC-4E; KSC LC-39A; CCAFS SLC-40
Total payload mass	619,967 kg
First successful ground-pad landing	2015-12-22
Successful drone-ship landings with payload 4,000–6,000 kg	4 booster records shown
Landing outcomes (2010-06-04 to 2017-03-20)	No attempt: 10; Success (drone ship): 5; Failure (drone ship): 5; Success (ground pad): 3; Controlled ocean: 3; Uncontrolled ocean: 2; Failure (parachute): 2; Precluded: 1

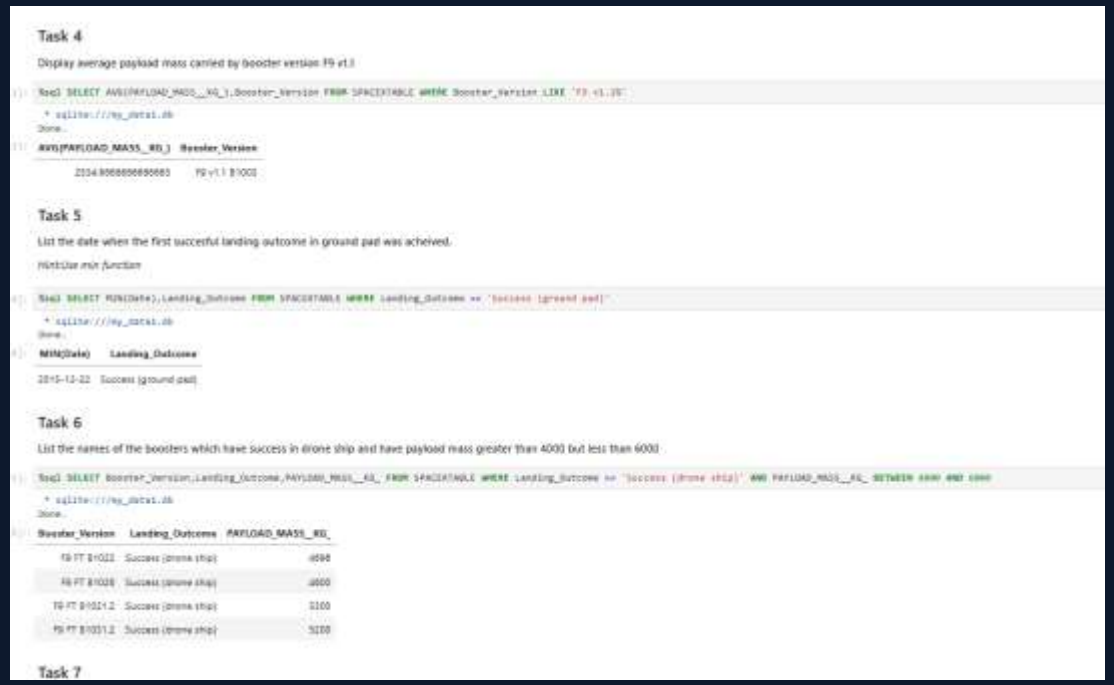
SQL: examples of reproducible analysis

Queries turn raw records into interpretable summaries

```
SELECT MIN(Date), Landing_Outcome
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

Payload mass total

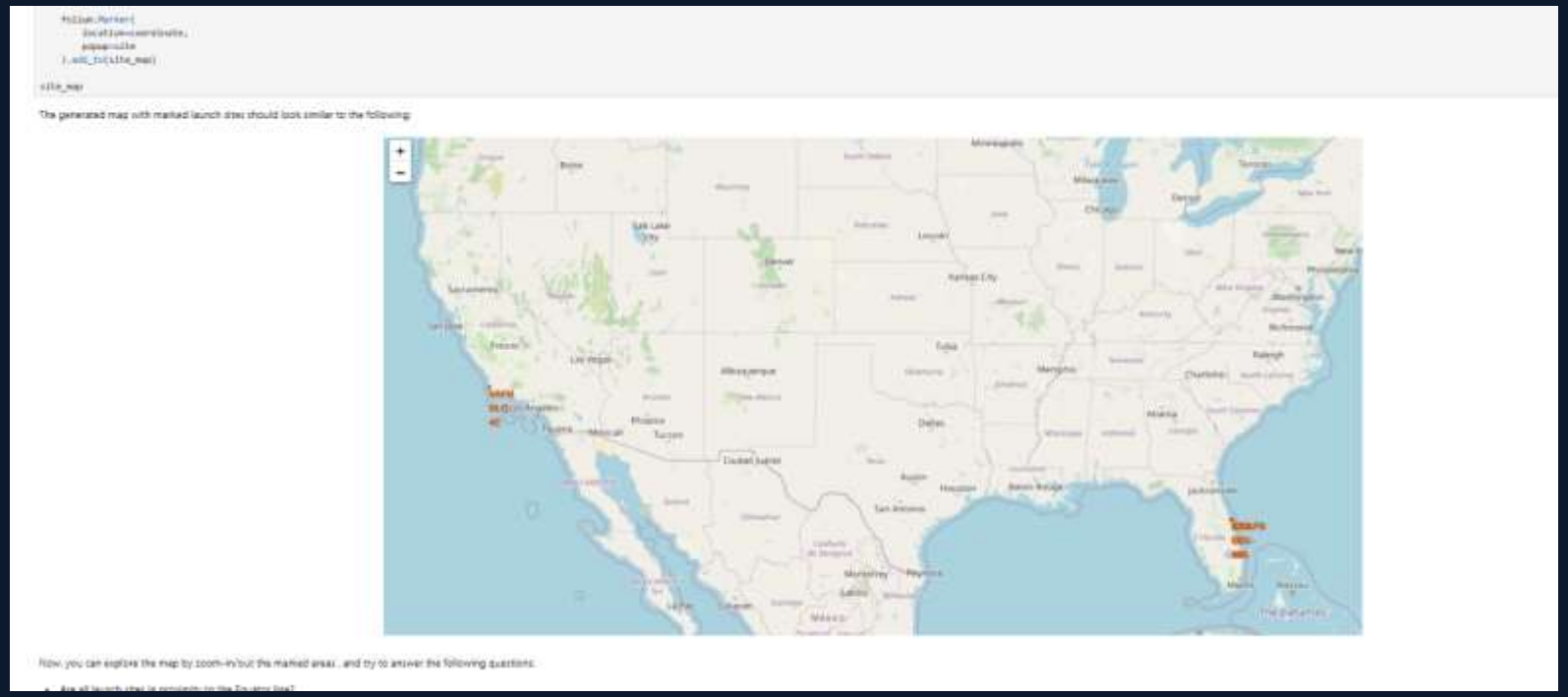
```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTBL;
```



Interpretation: SQL provides traceable answers, but query definitions and filters must be checked before drawing broader conclusions.

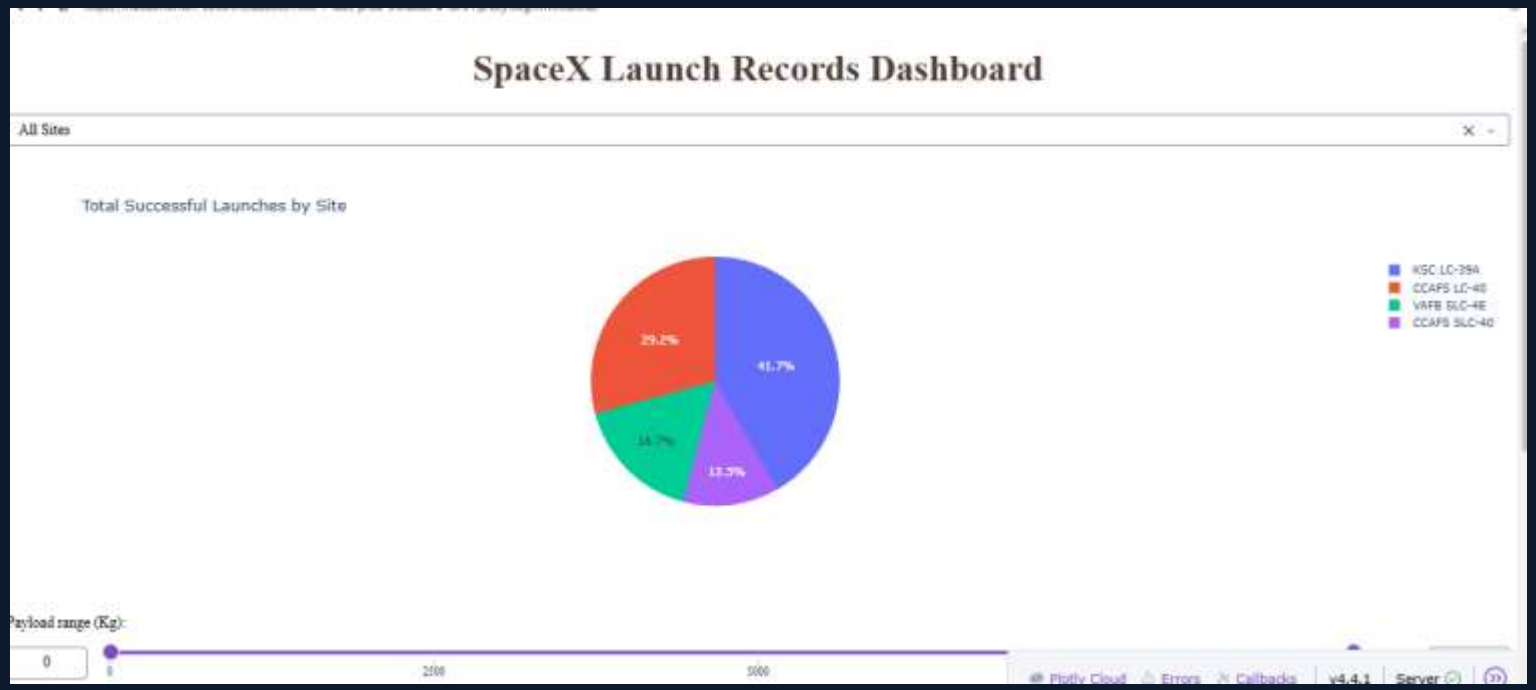
Geospatial analysis with Folium

Launch sites are concentrated on the U.S. West and Gulf/East coasts



Interactive analysis with Plotly Dash

A working dashboard connects filters to two linked views





- Filter by launch site.
- Adjust payload-mass range.
- Compare outcome and booster version.

Predictive analysis: classification

Cached notebook outputs — results should be revalidated on the corrected notebook

Model	Best CV accuracy	Test accuracy	Test correct
Logistic Regression	84.6%	83.3%	15 / 18
SVM	84.8%	83.3%	15 / 18
Decision Tree	89.1%	94.4%	17 / 18

Evaluation recommendation: include confusion matrices, precision/recall/F1, and a cross-validated comparison after rerunning the corrected pipeline.

Limitations & validation checks

What must be stated honestly in the final report

Conclusion & next steps

A practical, evidence-based summary

Core takeaway

The project demonstrates a complete analytical workflow—but model claims must remain proportional to the available evidence.

Repository & deliverables

Make the submission easy to review and reproduce

GitHub repository

https://github.com/MatiasMontenegrom/spacex-falcon9-landing-prediction

Screenshots in this deck were supplied by the project author; model scores are explicitly marked as cached and pending revalidation.

Data acquisition workflow

How the raw records become an analysis-ready dataset

Collection

- 1) Request historical launch records from the SpaceX REST API. 2) Read the course-provided Wikipedia launch table/scraped records. 3) Retain source URLs and the date of retrieval so the workflow can be reproduced.

Integration & cleaning

- Align launch identifiers and dates; select fields needed for the analysis; convert payload mass to numeric; parse dates; inspect missing values and duplicate records; normalize site and outcome labels before merging or aggregating.

Model-ready table

- Create the binary Class target from the course labels, keep it outside the predictor matrix X, encode categorical predictors, split train/test, then fit preprocessing on training data only. Check row counts and class balance after each transformation.

Model evaluation: what must be verified

The current metrics are cached; this slide distinguishes evidence from pending validation

Current evidence

- Logistic Regression: CV 84.6%, test 83.3% (15/18). SVM: CV 84.8%, test 83.3% (15/18). Decision Tree: CV 89.1%, test 94.4% (17/18). These are outputs cached from the earlier notebook, not confirmed results from the corrected pipeline.

Required final evaluation

- After executing the corrected notebook, regenerate a confusion matrix for each classifier and report precision, recall, F1 and accuracy on the same test set. Include cross-validation mean and variability where available.

Interpretation rule

- Do not select a model based on accuracy alone. With only 18 test cases, the uncertainty is large; one changed prediction moves accuracy by about 5.6 percentage points. Report the final test-set size and target-class distribution alongside the scores.